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Week 5

Banking Sector: Stress Test Framework

National Economic Council Financial Surveillance Training

27 August 2026

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Introduction

Banking stress test: A high-level definition

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What is a banking stress test?

"Bank stress testing is designed to test the **resilience of banks** to **severe but plausible shocks**. In practice, this typically means **modelling the impact** of hypothetical **adverse macro-economic and financial market scenarios** on bank profitability and balance sheets.

From the perspective of central banks and bank regulators, stress tests have the potential to **support both risk measurement and risk management**. For example, stress-test results help policymakers **set capita requirements**, which are in place to ensure banks fund themselves with **sufficient loss-absorbing capital** to reduce their likelihood of failure."

Source: Bank of England (2016)

Steps

Macro-scenarios

Macroeconomic stresses are designed to test the resilience of banks against the risks they face, such as severe recession combined with financial market distress. These scenarios are not based on economic projection; rather they represent unlikely severe outcomes chosen by stress testers because they could have a material detrimental impact on banks

Credit risk modelling and balance sheet transmission

Credit risk modelling estimates how macro scenarios can affect non-performing loans (NPLs) and credit provisions. These shocked NPLs and provisions will have impacts on profitability and capital adequacy via balance sheet transmission.

Interpretation and policy responses

Stress test results can be used for several purposes, with some authorities using them as a tool to highlight financial stability risks, some using them as part of their approach to setting individual bank capital requirements, and others using them to help set macroprudential policy as well.

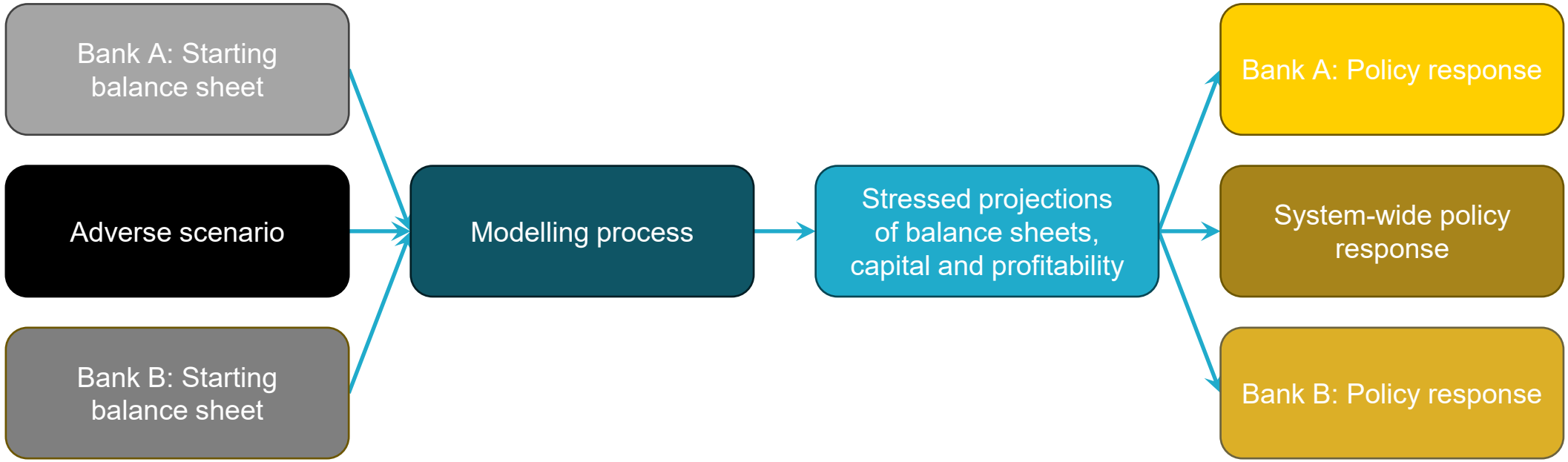
Guiding principles

2017 IMF FSAP Prudential Stress Testing

Credit risk modelling and balance sheet transmission

Calculation flow

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Stage 1: The adverse scenario is defined and the cut-off date for banks' starting balance sheets is specified.

Stage 2: A combination of banks' own models and those of the regulatory authorities are used to model the impact of the adverse scenario on banks.

Stage 3: At the end of the modelling process, a series of stressed projections are produced. These show the impact of the stress on banks' balance sheets, capital positions and profitability.

Stage 4: The stress-test results are used to inform both the system-wide and individual-bank policy responses. This could include changes to system-wide capital requirements, or improvements to banks' risk management processes.

What are macro-scenarios?

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The two key elements of this step are:

- to select the **types of risks** to be explored by the test
- to calibrate the **severity of shocks**.

Scenario design may simulate a **severe broad-based downturn** affecting the real economy as well as impacting financial markets and other asset prices.

- This approach allows stress-testing authorities to **factor in the correlation of different risks faced by banks**, and their impact across different parts of banks' balance sheets.
- Adverse scenarios of this type typically include **projections for economic variables** such as inflation, output growth and asset prices, which are broadly consistent with the paths these variables might be expected to take during severe downturns.

The complexity of macro scenario design may vary from simple macroeconomic shocks to complex system equation modelling (such as DSGE).

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Methods

Framework

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Banking stress test starts with macro-scenario modelling. These hypothetical macroeconomic scenarios are used to forecast future NPL and loan loss provisions (LLPs). Increments in LLPs will affect key financial metrics such as Earnings After Tax (EAT) and Capital Adequacy Ratio (CAR).



GDP growth
GDP reflects overall economic performance. Higher growth leads to better repayments.



BI 7-day repo rate
When a central bank raises interest rate to reduce inflation, the cost of credit increases.



Property index
Property prices represent consumer confidence and the performance of the real sector.



NPL prediction
Individual bank's NPL is predicted using dynamic panel data estimation based on macroeconomic scenario..



Calculation of provisions
Macro-scenarios, forward looking approach of provisioning follows the principles of IFRS 9's Expected Credit Losses (ECLs) principles.



Calculation of MtM securities losses
Using bond prices derived from changes in the policy rate, losses are calculated based on banks' holdings of FVTPL or FVOCI securities.



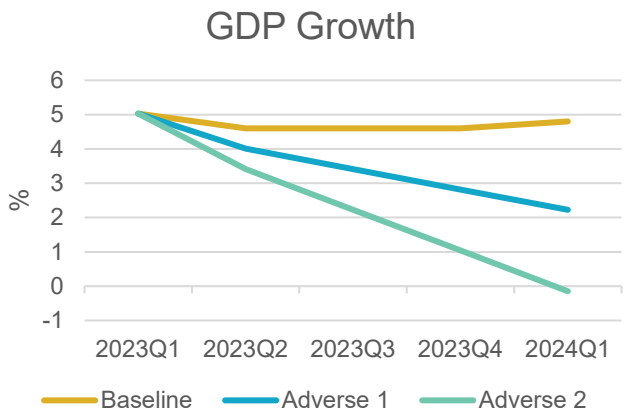
Surveillance and monitoring
Stress test results can be utilized to monitor the sufficiency of loss-absorbing capital and the probability of bank failure.

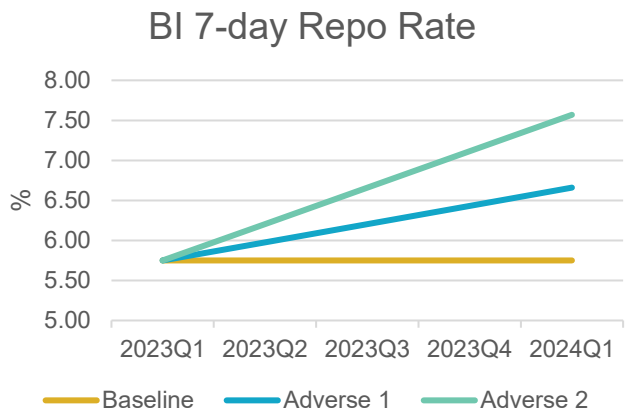


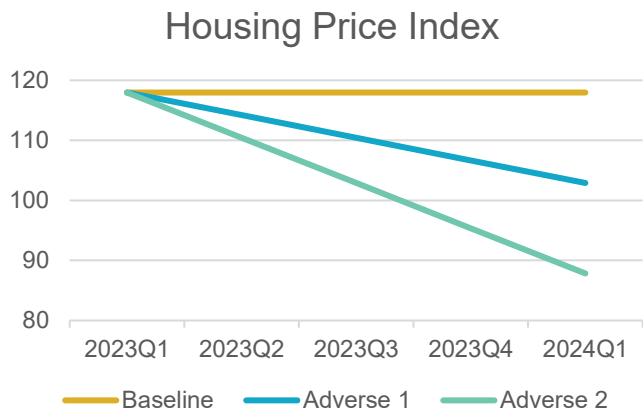
Policy responses
Policy responses and coordination such as the implementation of leverage ratio or bank resolution.

Macro-scenario generation

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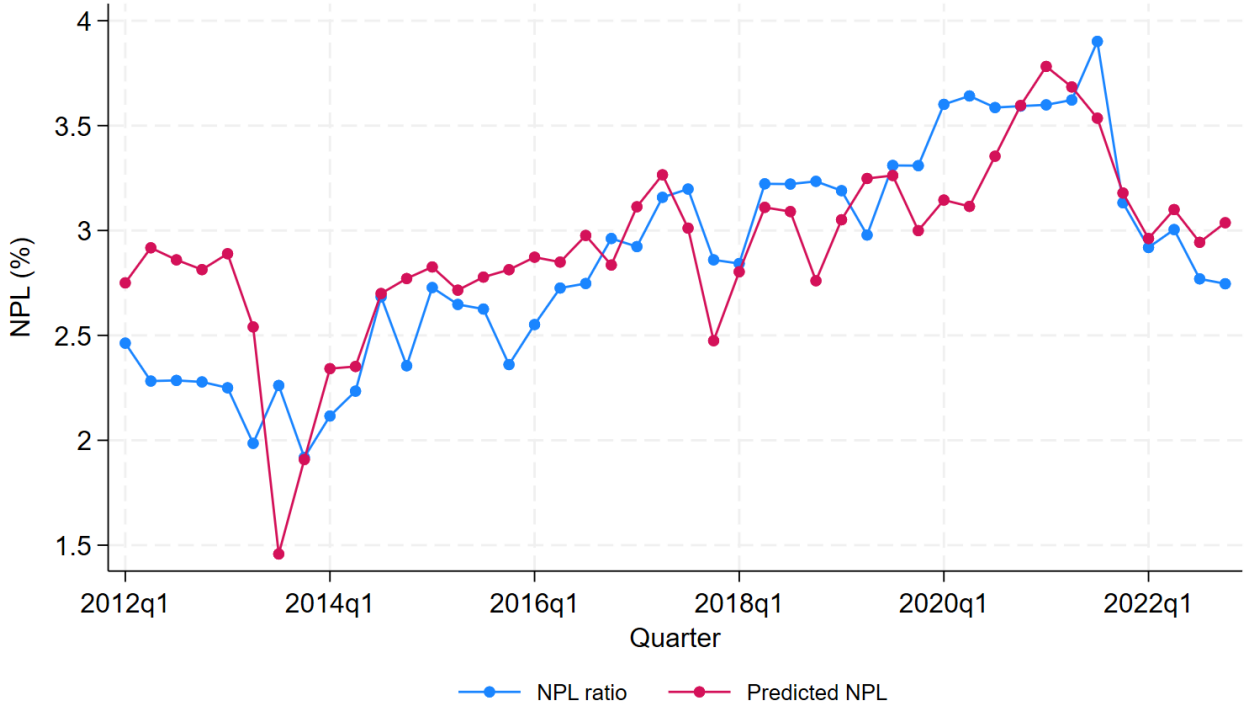


Scenario	Assumption
1. Baseline	Based on Prospera's macro forecast
2. Averse 1	0.25σ shock per quarter
3. Averse 2	0.50σ shock per quarter

Notes: Macro scenarios forecast four hypothetical quarters into the future.

Credit risk modelling (1)

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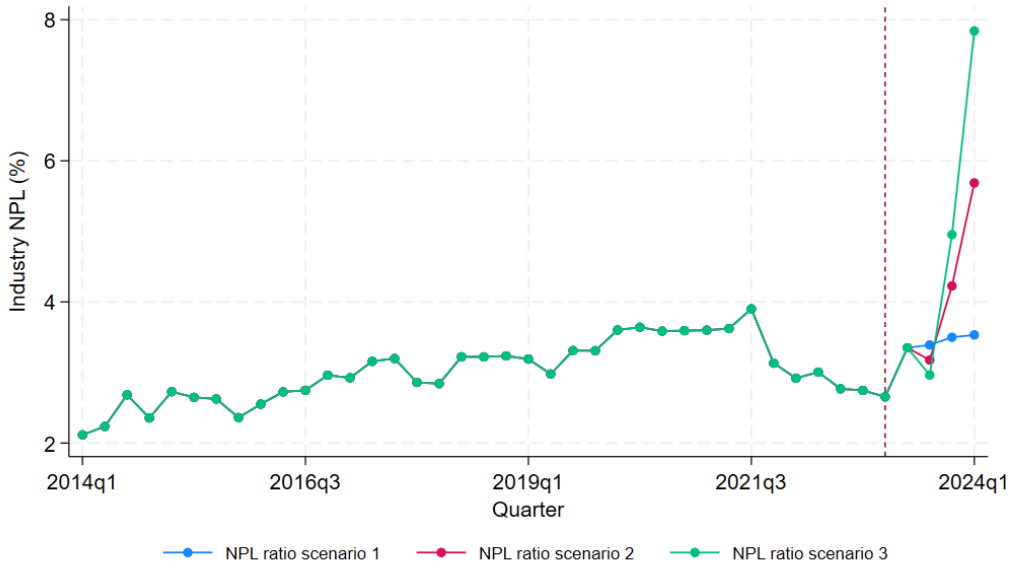


- We use quarterly 37 listed banks data between 2011 and 2022 to predict NPL trend using macroeconomic variables.
- Back-testing suggests goodness of fit of the NPL model.
- This justifies the use of macro-scenarios to forecast hypothetical NPL in the next four quarters.

Credit risk modelling (2)

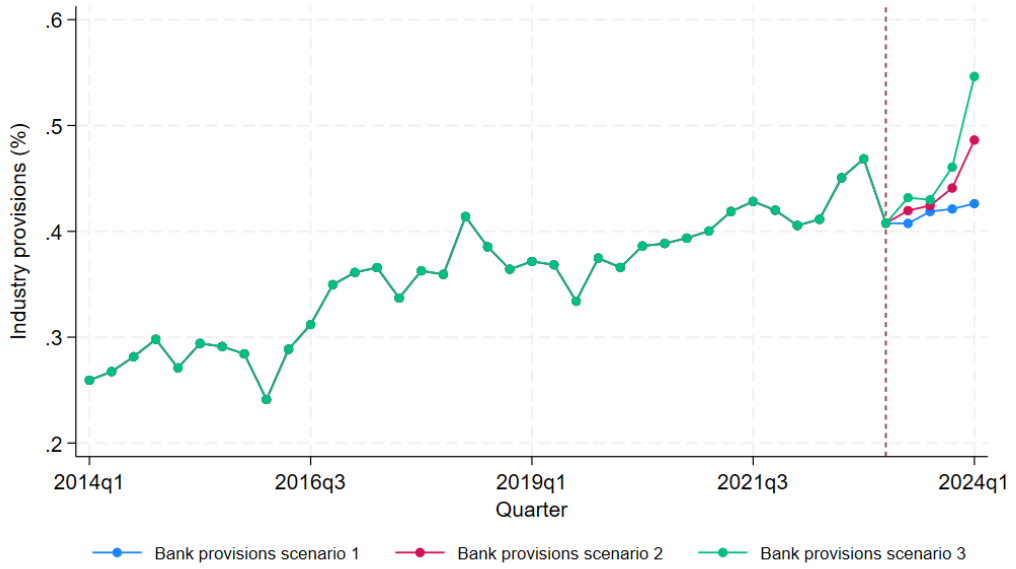
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NPL Forecast



- **Baseline scenario**
  NPL ratio remains stable around 3-4% until 2024Q1.
- **Adverse 1 scenario**
  NPL ratio increases to almost 6% by 2024Q1.
- **Adverse 2 scenario**
  NPL ratio increases to almost 8% by 2024Q1.

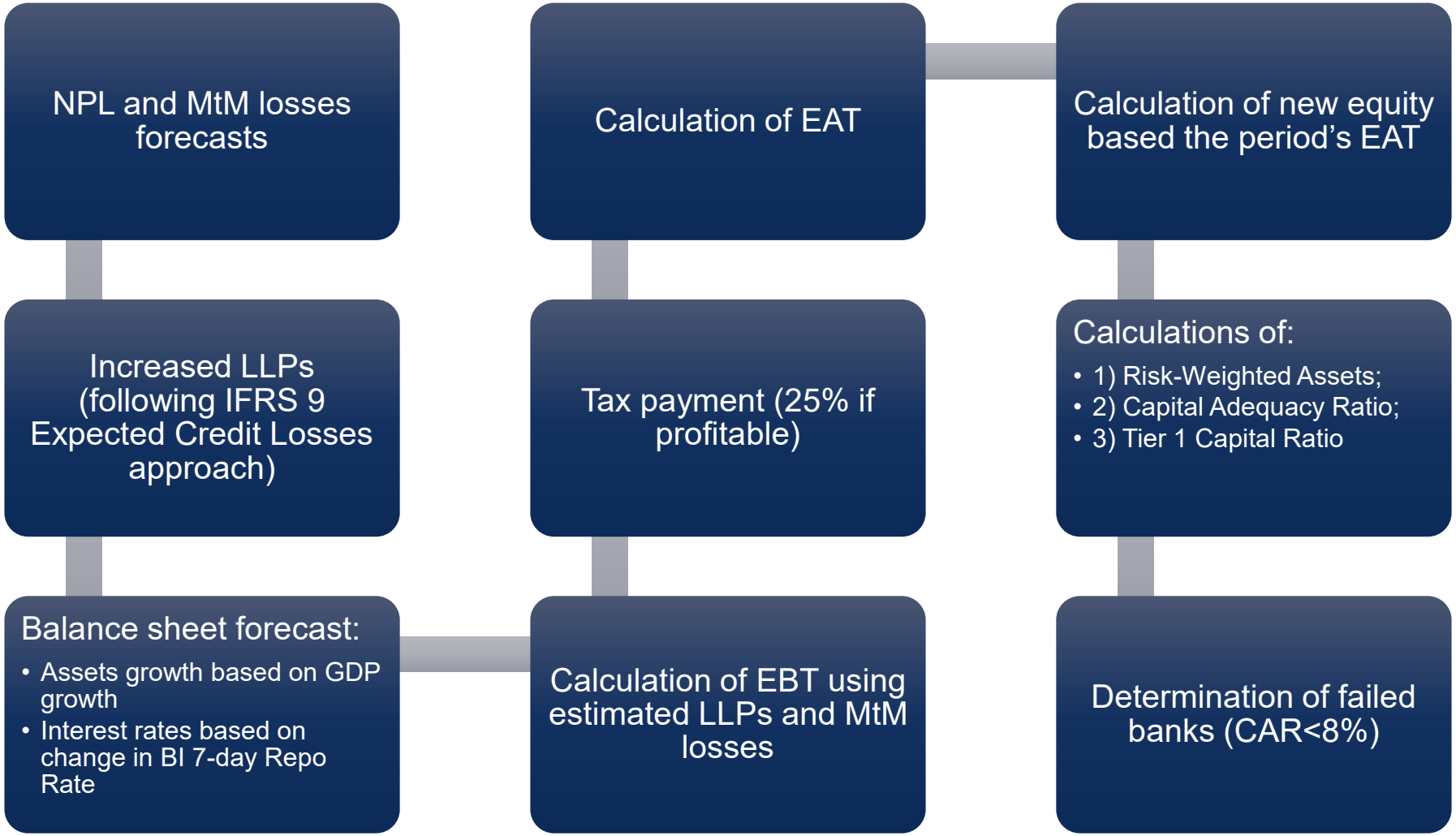
Provision Forecast



- **Baseline scenario**
  LLP-to-assets remains table around 0.4% until 2024Q1.
- **Adverse 1 scenario**
  LLP-to-assets increases to almost 0.5% until 2024Q1.
- **Adverse 2 scenario**
  LLP-to-assets increases to almost 0.6% until 2024Q1.

Balance sheet transmission

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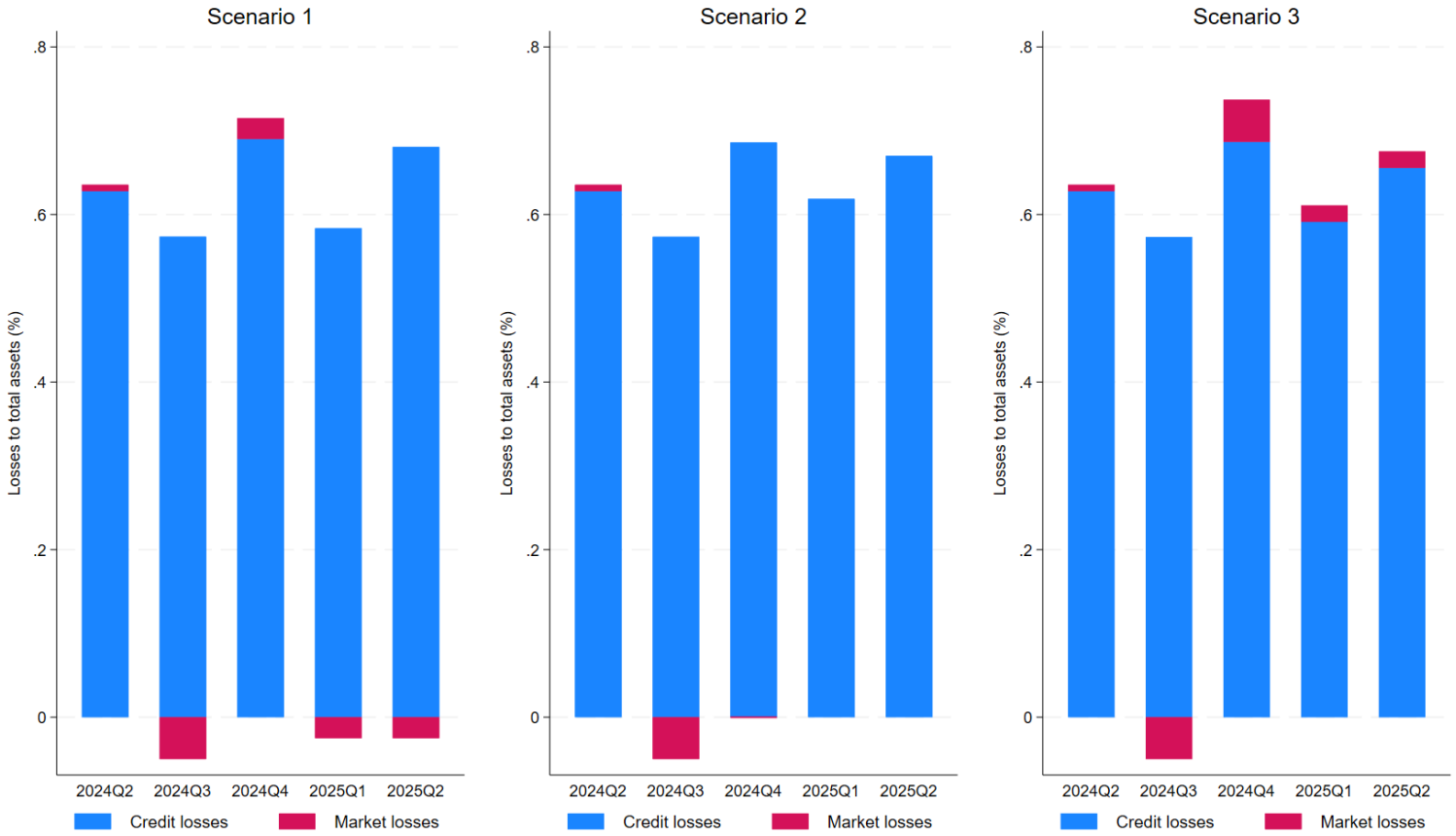


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Results

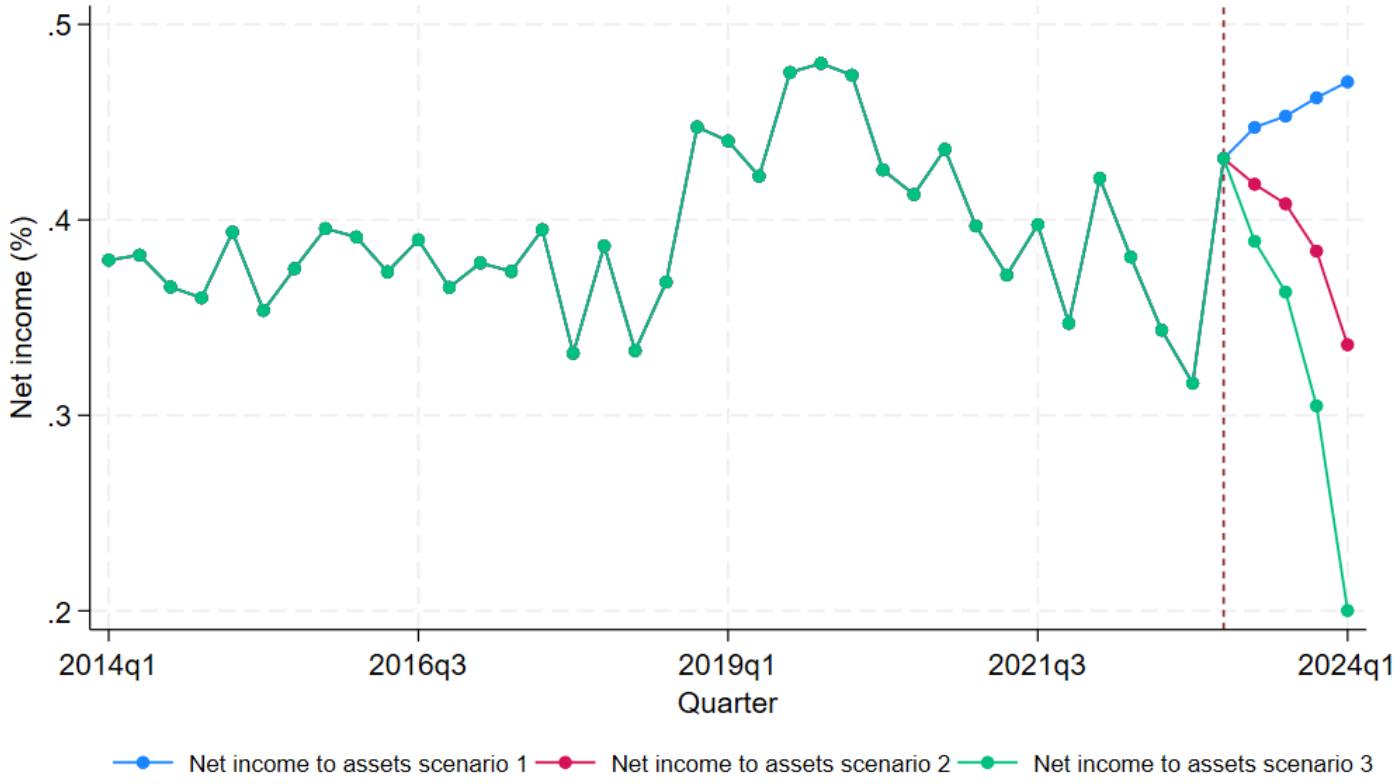
Composition of risks

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- Credit and market risks make up the losses that will affect the bank's solvency. The breakdown helps to identify the source of vulnerability.
- In some scenarios, market gains can help offset some of the losses from credit delinquency.

Profitability

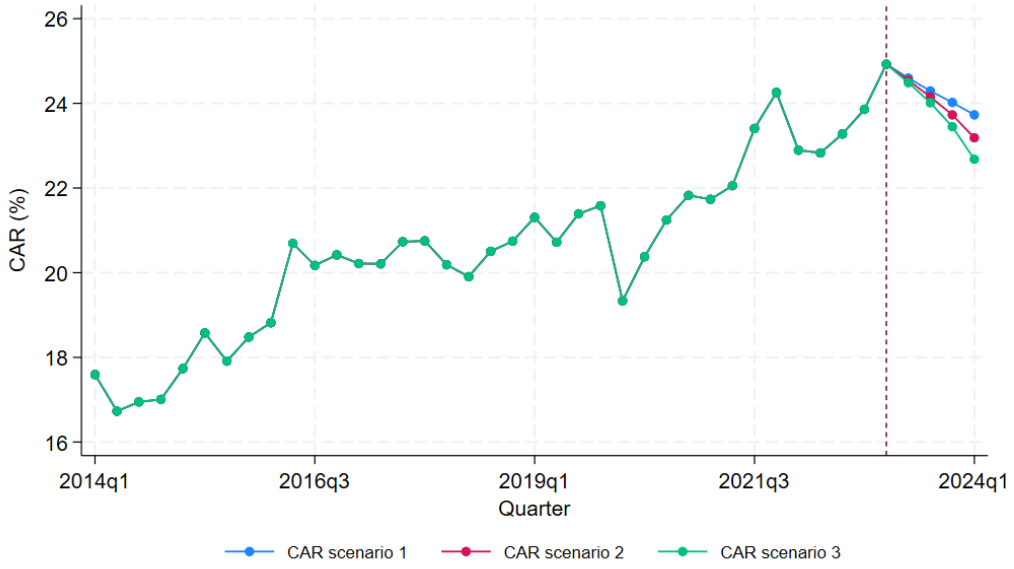


- Shocked LLPs lead to undermined banking sector profitability (EAT to assets), which falls to 0.2% under Scenario 3 (Adverse 2).
- Bank with lowest EAT after the shock is MCOR (Bank CCB Indonesia) with EAT to assets declines to -11%.

Capital ratios

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Capital Adequacy Ratio



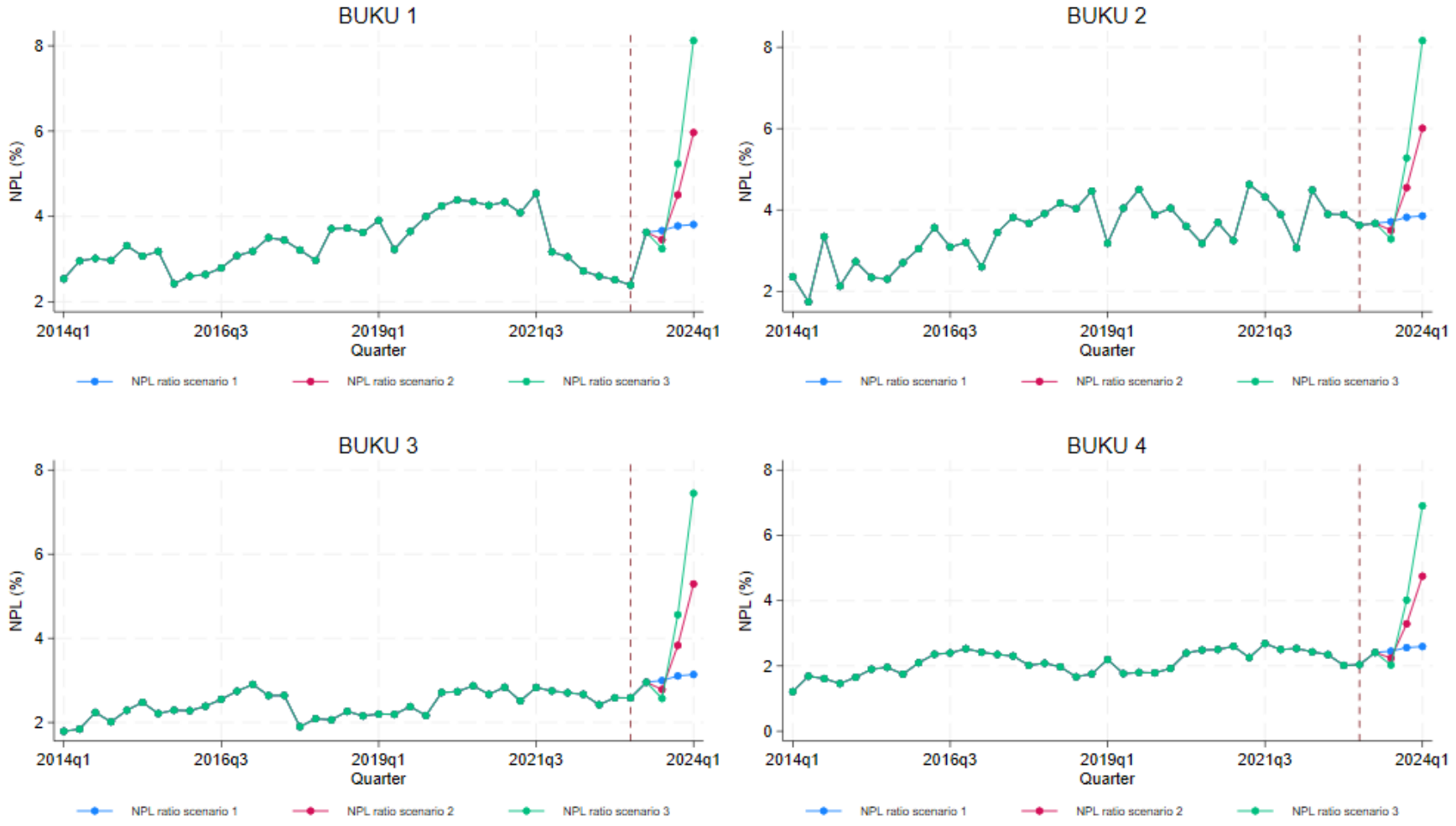
Tier 1 Capital Ratio



- Under Adverse 2 scenario, both CAR and Tier 1 Capital ratio fall by 2 percentage points by 2024Q1 from their latest points in 2024Q1.
- MCOR (Bank CCB Indonesia) is the only bank failing (CAR<8%) under both Adverse 1 and Adverse 2 scenarios, driven by its weak profitability.

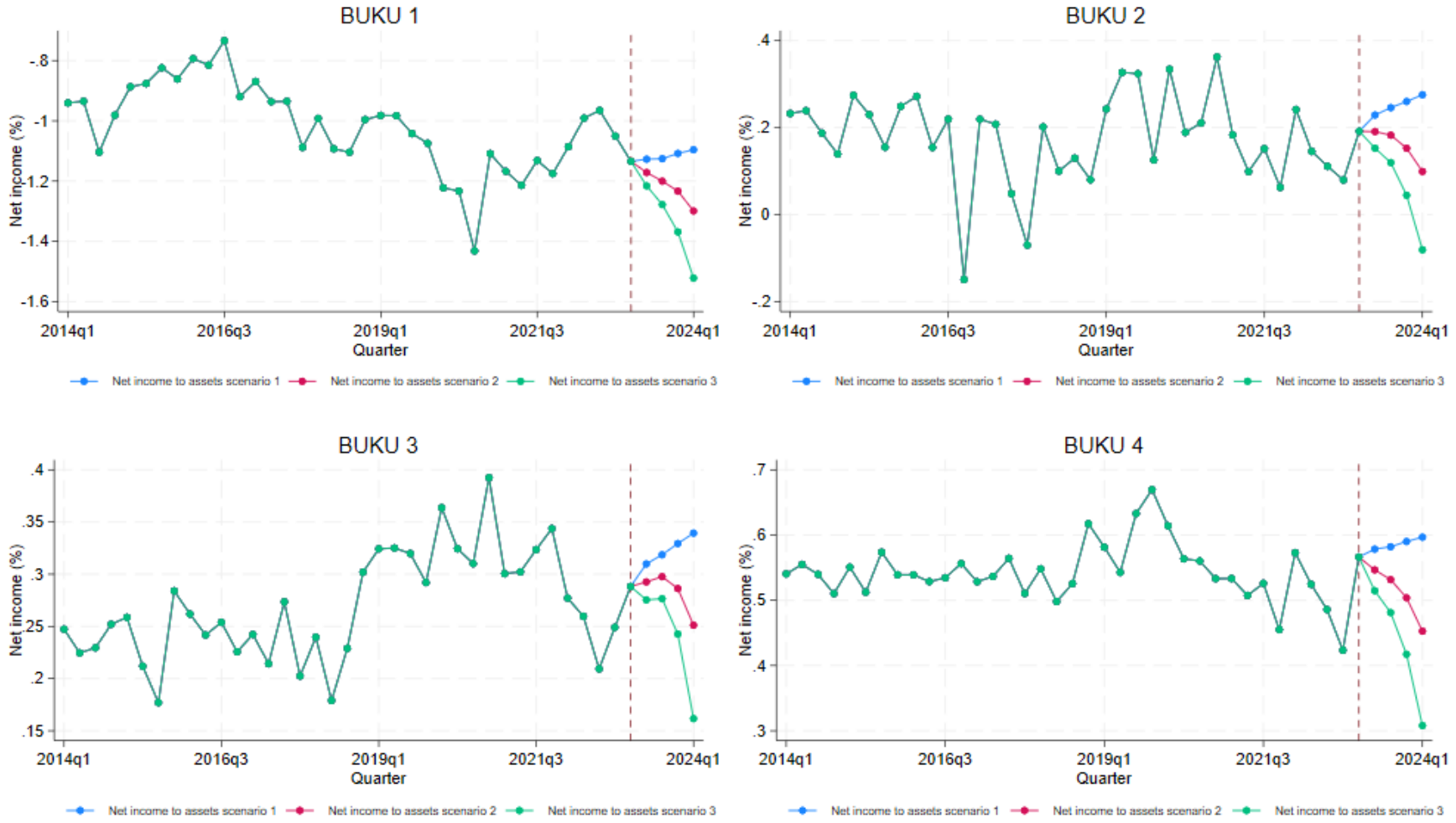
Non-performing loans by KBMI

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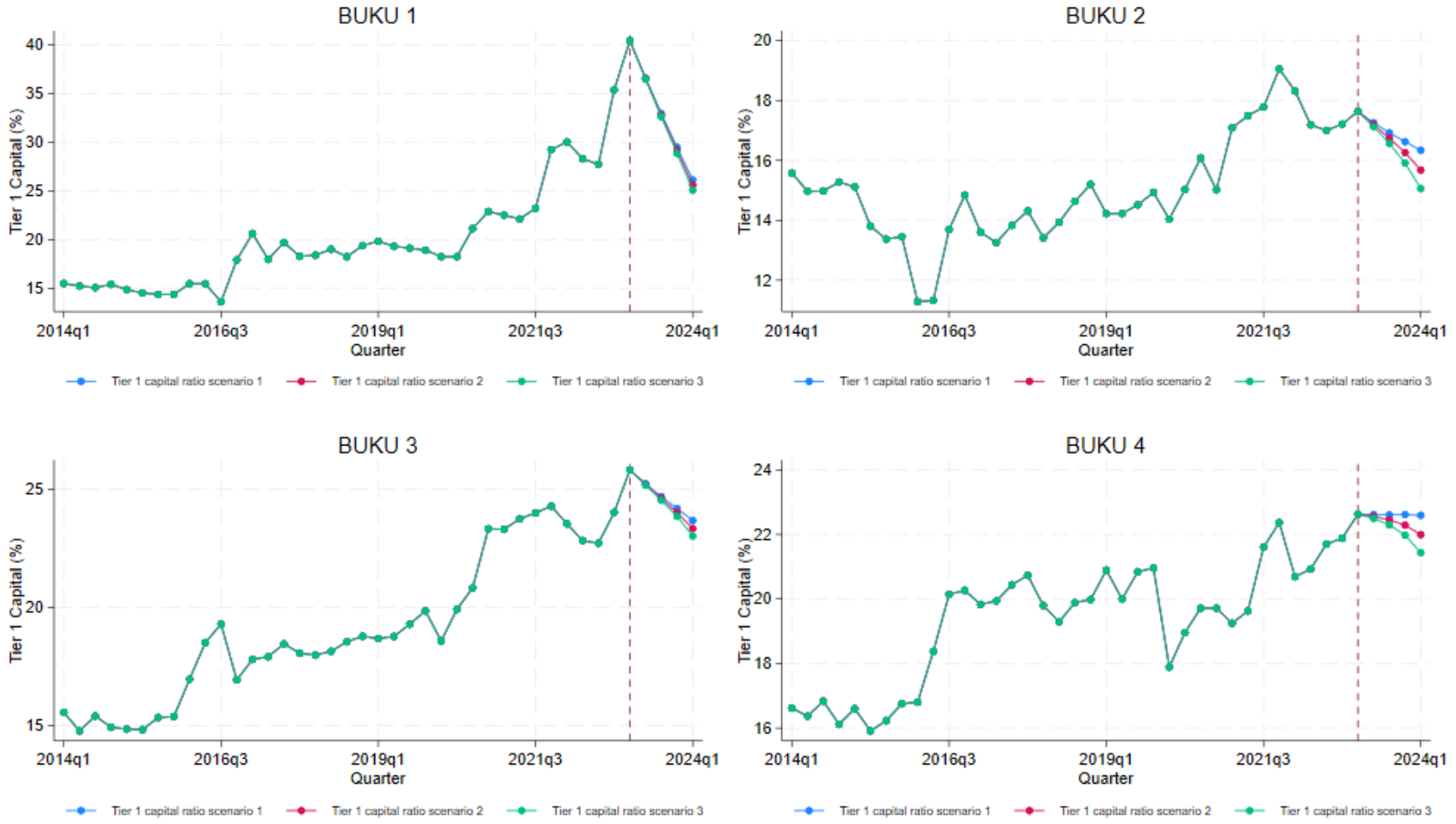
Profitability by KBMI

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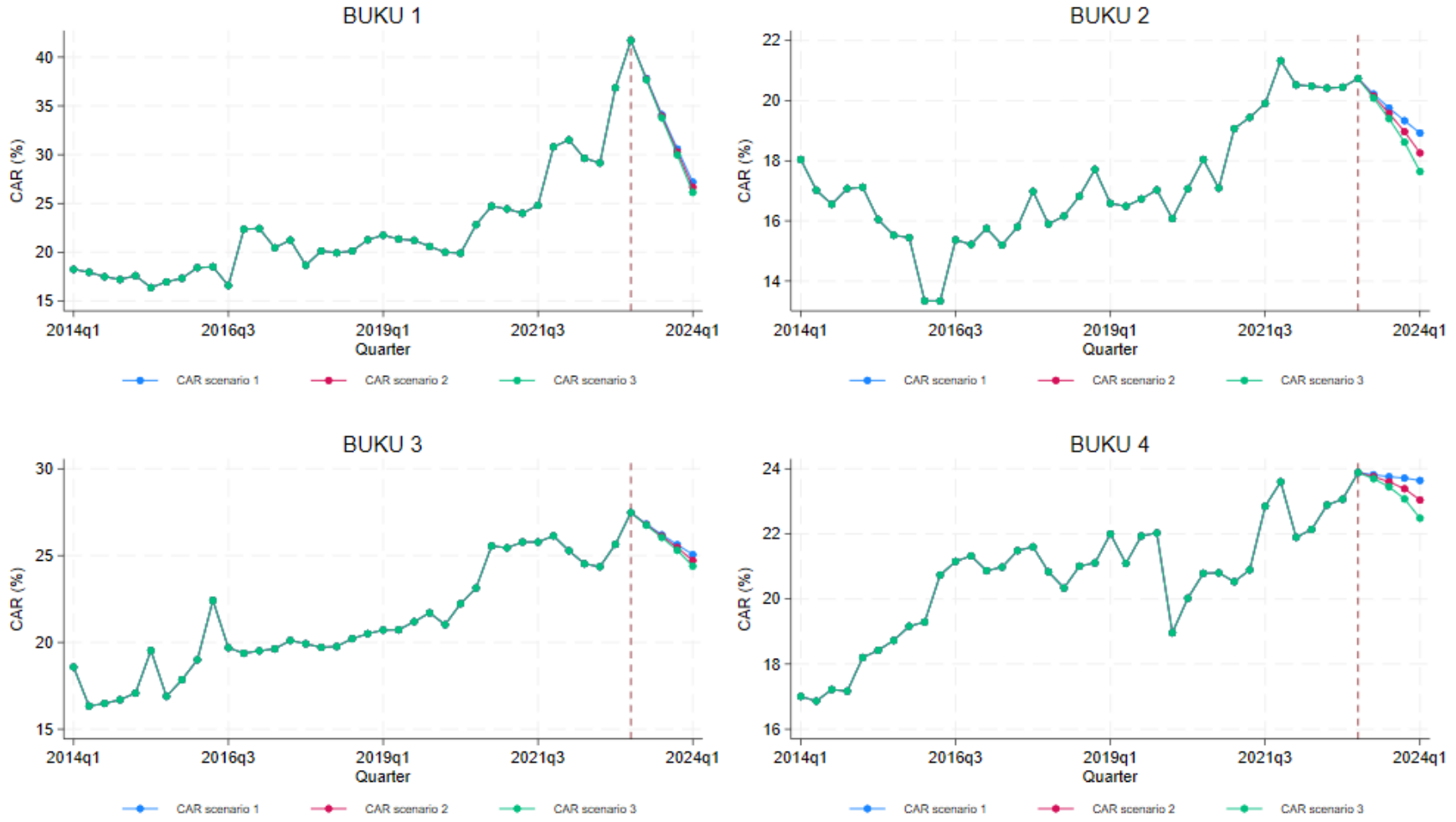
Tier 1 capital by KBMI

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Capital adequacy ratio by KBMI

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Feel free to ask any questions 😄

Thank You

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